

 exercise-3

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ID: SYS-01

Description: The user shall be able to interact with the system through a serial terminal

Rationale: To provide the user a simple interface to communicate with ESP32C6

Acceptance Criteria:

- The serial communication shall be configured as 115200-8-N-1
- The user shall be able to send commands from the terminal to ESP32C6.
- ESP32C6 shall correctly receive and process the commands and provide corresponding responses through the terminal.

ID: SYS-02

Description: The system shall be able to control a LED connected to ESP32C6.

Rationale: To provide control of the LED during system operation and testing.

Acceptance Criteria:

- The system shall be able to turn the connected LED on and off.
- The LED shall be connected to GPIO4.

ID: SYS-03

Description: The system shall prompt the user in the serial terminal to enter a command to control the LED.

Rationale: To inform the user of the available commands to control the LED.

Acceptance Criteria: The system shall display the exact prompt "Enter a command [0: Off, 1: On]:" in the serial terminal.

ID: SYS-04

Description: The system shall read and validate user inputs received through the terminal and it shall process only valid commands.

Rationale: To ensure that only valid user commands are processed.

Acceptance Criteria:

- The system shall accept valid commands
- Invalid inputs shall not trigger any system function.
- The valid inputs are '0', '1', LF and CRLF.

ID: SYS-05

Description: When the user presses Enter, the system shall prompt the user to enter a command.

Rationale: To provide a clear interaction mechanism that allows the user to enter the next command.

Acceptance Criteria: The system shall correctly detect both LF and CRLF as valid line terminators.

ID: SYS-06

Description: If the user enters '0' or '1' in the serial terminal, the system shall turn the LED off or on, respectively.

Rationale: To allow the user to control the LED through predefined commands.

Acceptance Criteria: The system shall respond to a valid LED control command without unnecessary delay.

ID: SYS-07

Description: The system shall display a success message in the terminal when a user command is executed successfully; otherwise, it shall display an error message.

Rationale: To provide the user with clear feedback about whether the requested command was executed successfully or not.

Acceptance Criteria:

- When a valid user command is executed successfully, the system shall display "=> done" in the terminal.
- When a valid user command cannot be executed successfully, the system shall display "=> failed" in the terminal.
- After displaying either message, the system shall display the command prompt again and allow the user to enter a new command.

ID: SYS-08

Description: The developed software modules shall support dual targeting, allowing them to be built and tested on both ESP32C6 and the development platform.

To enable this, the modules shall access ESP-IDF functionalities through an abstraction layer rather than directly depending on ESP-IDF APIs.

Rationale: To enable testing of the software modules on the development platform while maintaining compatibility with the target microcontroller.

Acceptance Criteria:

- Each developed module shall be buildable and testable on both the development platform and ESP32C6.
- Hardware- and ESP-IDF-specific functionalities used by the modules shall be accessed through a defined abstraction layer.
- The same module implementation shall be used for both targets, with only the abstraction-layer implementation differing where necessary.
- Unity shall be used as the testing framework.

ID: SYS-09

Description: CMake shall be used to configure, build and run the software tests.

Rationale: To provide a standardized mechanism for building and running the tests.

Acceptance Criteria:

- There shall be targets to build and run the tests on the development platform. The code-coverage results shall be generated in the html format.
- There shall be a target to build and run the tests on ESP32C6.

Application Layer: 03, 04, 05, 06, 07

Abstraction: Layer: 01, 02, 03, 04, 05, 06, 07

COM Stack: 01, 03, 04, 05, 07

IoHwAb: 02, 06

BSP: 01, 02, 03, 04, 05, 06, 07